

AIISH First Meeting Report

Project Title : Adaptive Augmentative and Alternative Communication (AAC) for Children with Cerebral Palsy (CP)

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Introduction

The first meeting at AIISH was conducted to gain a better understanding of the communication difficulties faced by children with Cerebral Palsy (CP) and to identify gaps in existing Augmentative and Alternative Communication (AAC) systems. The discussion focused on understanding the suitability of current AAC tools and defining the target users for our project.

During the meeting, Sahana Ma'am shared practical insights based on her experience working with children undergoing AAC training. The discussion helped us refine our project objectives and identify the specific needs of moderate Cerebral Palsy patients.

Discussion on Existing AAC Systems

Sahana Ma'am introduced us to some AAC methods and tools currently used for communication training. These included communication boards, picture-based communication systems, communication albums, and AAC applications such as Avaz.

Among these tools, Avaz is widely used for communication training and provides a large vocabulary set with multiple communication options. However, Sahana Ma'am highlighted that the application can be difficult for many children with Cerebral Palsy because of its complexity. The large number of options, categories, and navigation steps can make communication slower and more challenging.

The discussion also included devices such as Buddy Button, which provide simple switch-based communication support. While these devices are useful for basic communication, they have limited flexibility and vocabulary compared to modern AAC applications.

Understanding the Target User Group

Initially, there was a discussion regarding the motor skill limitations of Cerebral Palsy patients. Based on guidance from our faculty members, we raised concerns about whether motor impairments would affect the usability of AAC systems.

Sahana Ma'am clarified that our project should focus specifically on children with moderate Cerebral Palsy. These children generally possess some level of motor control and can interact with digital devices. Therefore, the primary challenge is not the complete lack of motor skills but the complexity of existing communication systems.

Moderate CP users often face difficulties when:

- * Navigating through multiple layers of menus.
- * Searching for appropriate symbols or words.
- * Handling large and complex vocabularies.
- * Using systems that require extensive learning and training.

Therefore, our project should focus on reducing complexity rather than designing for users with severe physical limitations.

Key Insights from the Meeting

The major observations and suggestions provided by Sahana Ma'am are as follows:

1. Avaz is Too Complex for Many Users

Although Avaz is a powerful AAC application, it is more suitable for users with mild impairments who can comfortably navigate through multiple options. Moderate CP users often find it overwhelming and difficult to use independently.

2. Need for a Simpler Interface

The proposed solution should provide a clean and intuitive interface that minimizes the number of choices presented to the user at a given time. The goal is to reduce cognitive load and make communication faster.

3. Adaptive and Dynamic Communication System

One of the most important suggestions was to develop an AAC system that adapts to the user.

The application should:

- * Learn from frequently used words and phrases.
- * Prioritize commonly selected communication options.
- * Dynamically adjust according to user behaviour.
- * Reduce navigation effort by displaying relevant choices.

4. Retain Existing AAC Communication Methods

Current AAC approaches such as pictures, symbols, communication boards, and communication albums are already familiar to children during therapy sessions. Therefore, these communication methods should be retained within the proposed system rather than replaced.

5. Improve User Independence

The ultimate goal is to help children communicate with minimal assistance from caregivers. A simpler and adaptive system can encourage independent communication and improve participation in social and educational activities.

Problem Statement Refinement

Based on the meeting, we refined our problem statement as follows:

Existing AAC applications such as Avaz provide extensive communication support but are often too complex for children with moderate Cerebral Palsy. These users can operate digital devices but face difficulties navigating large vocabularies and multi-level interfaces. As a result, communication becomes slow, frustrating, and dependent on caregiver assistance.

There is a need for a simplified, adaptive AAC application that retains familiar picture-based communication methods while dynamically adjusting to user preferences and communication patterns.

Proposed Project Direction

The proposed solution will focus on developing a user-friendly AAC application with the following characteristics:

- * Simplified navigation structure.
- * Picture and symbol-based communication.
- * Dynamic adaptation based on frequently used content.
- * Reduced cognitive complexity.
- * Faster message generation.
- * Improved accessibility for moderate CP users.
- * Support for independent communication.

Conclusion

The first meeting at AIISH provided valuable insights into the practical challenges faced by children with moderate Cerebral Palsy when using existing AAC systems. While tools such as Avaz offer extensive communication capabilities, their complexity often limits usability for the target user group. Based on the guidance provided by Sahana Ma'am, our project will focus on designing a simplified and adaptive AAC application that enables easier navigation, faster communication, and greater independence for children with moderate Cerebral Palsy



